PhishGuard: a fine-tuned NVIDIA Nemotron beats a frontier model at scam screening

Nemotron 3.5 Super + LoRA on Fireworks
Email · SMS · URLs · phone-call transcripts · 600 held-out cases

97.8% vs 96.2% for Kimi K3 on 600 held-out cases (p = 0.041)

~7× cheaper · 23× smaller · open model

Team: John · Kris | github.com/yuansshi/fireworks_hackthon

Scams hit every channel. Frontier screening costs too much.

4

channels to screen:
email, SMS, links, calls

$1,600

per 1M messages with a frontier model (Kimi K3)

Now

a verdict is needed before the click or the call ends

User: fraud and security teams at banks, telcos and enterprises that must screen every message.

One evaluation, frozen before any run

2,400
training examples
(600 per channel, balanced)

600
held-out test cases
(150 per channel)

4
channels, one
malicious / benign label

Plus: 200 emails from an unseen later corpus (2008) · 200 recent synthetic scams · 300 call prefixes

Same hash-checked files and scoring for every system · data from Hugging Face

Four systems, the same cases, the same prompt

Kimi K3 (2.8T)	Frontier baseline; metadata + timing cues in the prompt
DeepSeek V4.1 Flash (552B)	Cost-efficient frontier baseline
Nemotron 3.5 Super, untuned	The "before": same prompt, same decoding
Nemotron 3.5 Super + LoRA	Ours: trained on the 2,400 examples; prompt unchanged

The lightweight training run

Rank 16

LoRA adapter on Nemotron 3.5 Super (120B / 12B active)

1 epoch

75 steps × 32 examples, learning rate 1e-4

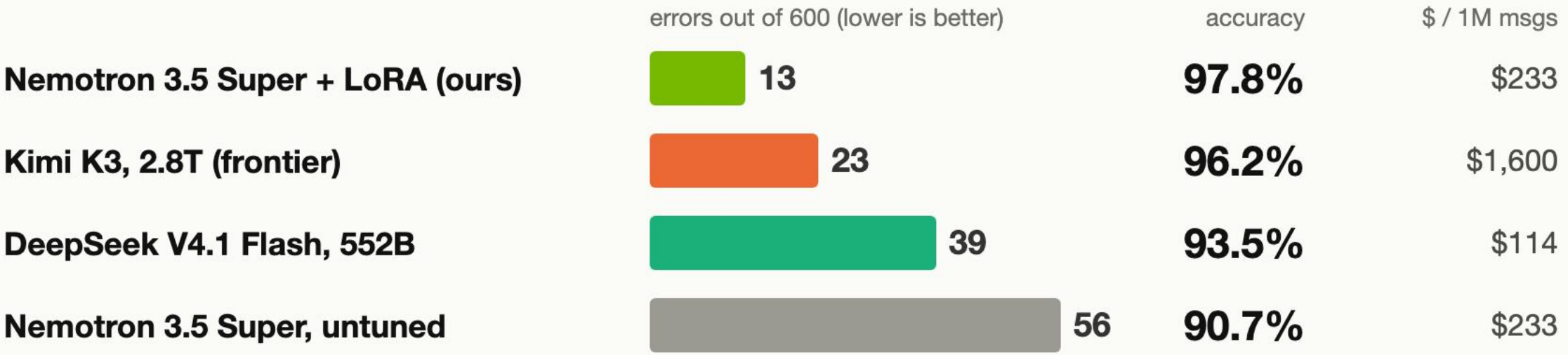
~11 min

training time

Training charge for the evaluated epoch: ~$2.20 (944k tokens × $2.32 / 1M) · whole project ≈ $8 of $50

Fireworks Serverless Training API: no dataset upload, no deployment; base and tuned adapter sampled in-session

Held-out accuracy and cost



+10 net correct vs Kimi K3 (15 vs 5 discordant), McNemar p = 0.041 · ΔF1 +1.7 [+0.2, +3.3]

43% fewer errors than Kimi K3 · +43 correct vs the same model untuned · about 7× cheaper

Demo: held-out items, ours vs. the frontier

```
$ uv run python scripts/demo.py --models ft-nemotron-super kimi-k3

[SMS]  truth: MALICIOUS
  Here is your discount code RP176781. To stop further messages reply stop…
  ✔ ours    MALICIOUS   recorded   $156 / 1M msgs
  ✔ Kimi K3 MALICIOUS   1.33 s     $1,218 / 1M msgs

[EMAIL]  truth: benign
  Subject: RE: LOCATION FOR MAN NIGHT  Either way, it sounds like…
  ✔ ours    benign      recorded   $518 / 1M msgs
  ✔ Kimi K3 benign      0.91 s     $2,973 / 1M msgs

[URL]  truth: MALICIOUS
  https://www.citieid.com/kenya
  ✘ ours    benign      recorded   $153 / 1M msgs
  ✘ Kimi K3 benign      0.69 s     $1,212 / 1M msgs

[CALL, first 50%]  truth: MALICIOUS
  caller: Hello, this is Karen from the Microsoft Refund Department…
  ✔ ours    MALICIOUS   recorded   $275 / 1M msgs
  ✔ Kimi K3 MALICIOUS   1.45 s     $1,836 / 1M msgs
```

Ours: verdicts recorded in the evaluation run (the adapter was sampled in its training session) · Kimi K3: live API calls

Timing: early warning, and the honest tradeoff

After hearing 25% of a call

Ours: scams caught	**96%**
Ours: false alarms	**2%**
Kimi K3: scams caught	98%
Kimi K3: false alarms	14%

From 50% of the call on, ours: 100% caught, 0% false alarms

Drift: where the frontier wins

	ours vs Kimi K3
Unseen 2008 email corpus, F1	83.8 vs **91.5**
Recent synthetic scams, accuracy	86.0 vs **94.0**

Next: refresh the adapter on recent data and route out-of-distribution traffic to a frontier fallback.

Why adopt PhishGuard

97.8%

beats the frontier on
the target task (p = 0.041)

~7×

cheaper per message
than Kimi K3

23×

smaller model
(120B vs 2.8T)

An open NVIDIA Nemotron model you can self-host, so customer messages stay on your infrastructure.

Reproduce everything: **github.com/yuansshi/fireworks_hackthon** (make data · make report)